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Equivalent circuit model of the carrier-depletion-based push-pull silicon optical modulators with T-rail slow wave electrodes

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***Abstract*—The t-rail electrode is an effective method to enhance the silicon optoelectronic modulator's performance. To design and optimize T-rail electrodes, engineers often rely on finite-element numerical simulations that require complex device modeling and enormous computing resources. In this paper, we present an equivalent circuit model for carrier-depletion-based push-pull silicon modulators with T-rail electrodes. The analytical solution for the bandwidth of the modulator can be derived from the equivalent circuit. The utilization of the analytical solution offers advantages in terms of memory conservation and flexibility. The values calculated by the equivalent circuit model are in excellent agreement with the numerical full-wave HFSS simulations. Hence, the proposed model can accurately and efficiently develop silicon optical modulators.**

***Index Terms*—Traveling wave devices, optical modulation, equivalent circuits, optical communication equipment, microwave propagation.**

I. INTRODUCTION

WITH the ever-growing need for broadband communications and the gradual and unwavering advancement of optical technologies, the demand for optical modulators is increasing at a dizzying rate. Silicon photonics (SiP) is extremely tempting for short-reach optical interconnects due to its complementary metal oxide semiconductor (CMOS) compatibility. The silicon modulator is a critical component of SiP communication connections [1]. Data transmission and reception rely heavily on high-bandwidth electro-optic (EO) modulators. An ideal modulator should possess a wide bandwidth, low half-wave voltage ($V\pi$), and be compatible with standard CMOS technology.

The maximum bandwidth of modulators based on silicon is currently 110GHz [2]. The slow light effect in a coupled-resonator optical waveguide (CROW) architecture is utilized. The device efficiency factor is 0.013 π/V which indicates an extremely large $V\pi$ of 77V. Although with a record-breaking bandwidth, this modulator design requires special production methods and possesses considerable $V\pi$. A substrate-removed 2mm-long MZM with a modulation bandwidth exceeding 50 GHz was reported ,The modulator made by changing the substrate material has a suitable $V\pi$ of 7V [3]. However, this type of structure hinders the fabrication process and increases the sensitivity of the device to mechanical vibrations. Based on compatible and common CMOS technology, improving the performance of modulators is usually achieved by changing the electrode structure and introducing a slight impedance mismatch. Two traditional traveling wave electrodes (TWE) are coplanar waveguides (CPW) and coplanar strip lines (CPS) [4-9]. The radio frequency (RF) index, characteristic impedance, and conductor loss are all influenced by variations in the width, gap, and thickness of CPS and CPW. It is unable to meet the requirements of high refractive index, impedance matching, and low loss simultaneously [4, 9]. Hence, the performances of the silicon photonic modulators based on CPS and CPW are limited. The reported bandwidths of silicon optoelectronic modulators based on these two electrode configurations are both generally less than 40 GHz [10-15]. For example, At a bias voltage of 0.5V, for a 2mm modulator based on CPS has a relatively low bandwidth of only 10G and a relative low $V\pi$ of 2V[16]. Another 3.3mm-long modulator based on CPS possesses a smaller bandwidth of 15G and a smaller $V\pi$ of 3.35V [11]. A 2mm long modulator using CPW electrodes can achieve a bandwidth of 20G and a $V\pi$ of 6.4 V [17]. The 4.2mm-long modulator using CPS and segmented PN junction has a high bandwidth of 35G and a $V\pi$ up to 7.5V [14]. Despite substantial attempts to improve the performance of silicon modulators, such as shortening the device length to 1mm [7] and implementing an impedance mismatch [6], the electro-optic (EO) bandwidth of modulators based on CPS and CPW has been limited to a maximum value of 46 GHz [6]. The 2.5mm-long modulator is based on CPS and a slightly

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impedance mismatch and has a Vπ of 7.6V. The use of T-rail slow wave electrodes (TSWE) has been empirically demonstrated to enhance the refractive index of microwaves, thereby leading to an expansion in the bandwidth of optical modulators [8, 18-22]. In the same device length, the T-rail electrode modulator achieved a bandwidth of 41 GHz, which is substantially higher than that of the modulators based on CPS and CPW [8]. The 4.25mm long modulator using T-rail electrodes has a 41G load and a Vπ of 8V under 8V bias voltage. Based on this T-rail device design scheme, a substantial bandwidth greater than 67GHz was successfully achieved by shortening the device length and adopting a segmented structure. T-rail electrode modulators produced under general technology can possess a bandwidth beyond 67GHz and Vπ of 5V [23]. The T-track electrode arrangement is an increasingly common choice for high-performance modulators.

To develop a high-performance modulator, a design method for a suitable t-track traveling wave electrode is crucial. The application of finite element method (FEM) simulations is a common method for reaching a high degree of accuracy in complex 2D or 3D device modeling. Nevertheless, this approach exhibits a weakness in terms of efficacy due to its protracted duration for numerical computation and substantial demand for computational resources. Furthermore, the comprehension of simulation outcomes might be difficult in the absence of an established physical structure. Another method entails the creation of an equivalent electrical circuit that aligns with the physical architecture of the modulator. The utilization of an equivalent model offers significant benefits owing to its lower computational demands in comparison to FEM [9, 24-26]. Furthermore, it provides valuable insight into the physical characteristics of the transmission line. Although the viability and efficiency of this method of similar circuit analysis have been demonstrated on CPS and CPW [24, 25, 27, 28], there is a lack of corresponding research on T-rail electrode structures. Thus, the equivalent circuit model for the T-rail electrode is essential for designing silicon modulators.

In this paper, straightforward closed-form equations for the propagation parameters of TSWE are provided. Various variables are considered in the design process, including electrode thickness, asymmetric schemes, and two-layer substrates, among others. The theoretical capacitance of the diode is incorporated into the capacitance of the TSWE to assess the overall impact. The modulator parameters calculated by using the proposed equivalent circuit model are highly consistent to the HFSS simulation. Due to the accuracy and usability of this circuit model, it is believed that these formulas possess the ability to serve as valuable references for designs.

II. Equivalent circuit model

A. Transmission line

Fig. 1(a) shows the schematic of the modulator that consists of a traveling-wave electrode and a carrier-depletion-based p-n junction. The electrode is a ground-signal (GS) metal periodically loaded by T-shape capacitors. Figs. 1(b) and 1(c) show the top view and the cross-sectional view of the modulator. A lateral p-n junction is embedded in the center of a rib waveguide in this arrangement. While highly doped P++ and N++ regions are employed for ohmic connections. The fixed device parameters used in the simulation are h_{sub}= 750μm, h_{box}= 2μm, h_{clad}=1.5μm, h_{slab}=90nm and t=2μm. The modulator was fabricated in a high-resistivity handle wafer (>750 ohm-cm). In this case, the space charge area exhibits a minor dielectric loss tangent at the operating frequency because of its light doping. As a result, silicon in the region of PN doping functions as a dielectric. The silicon is permeable to both transverse magnetic and transverse electric fields. "Dielectric quasi-TEM mode" is the propagation mode. In the slow-wave mode guided by a micron-sized electrode, quasi-TEM analysis is valid since the very small cross-section of the transmission line ensures that the transverse fields are essentially quasi-stationary [23,26–28]. As the circuit parameters generated during microwave propagation can be calculated by conformal mapping, partial capacitance method, and other methods, TSWE can be described by equivalent circuits.

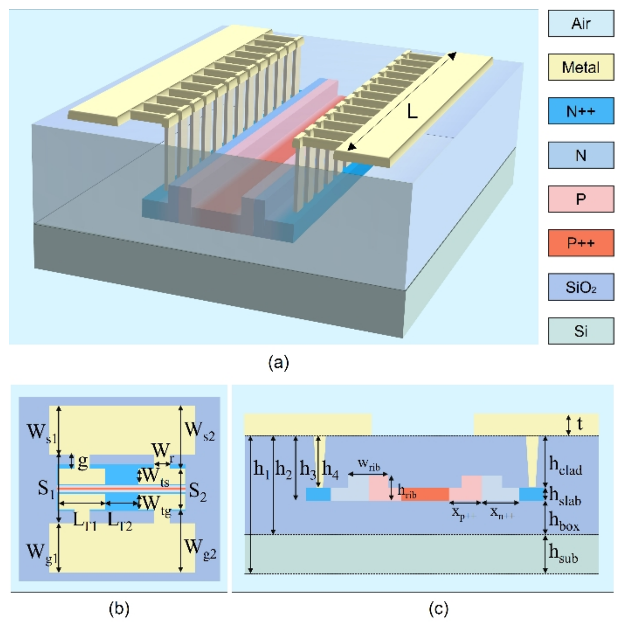

Fig. 1. (a) Overview of the depletion-based silicon optical modulator with segmented electrode. (b) Top-down detail view of the electrode. (c) Cross section of a typical carrier-depletion-based optical modulator in SOI with the TSWE (drawing not to scale).

Periodic T-rail loading mainly increases the capacitance per unit length [8, 22, 26]. The equivalent circuit model is shown in Fig. 2(a). R_{trail} and C_{trail} are the resistance and capacitance per unit length due to the T-rail. C_j and R_j represent the depletion capacitance and series resistance per unit length of the p-n junction. The remaining components align with the CPS transmission line. Fig. 2(b) shows the relationship between the circuit and the physical device. The above constituent circuit components can be transformed into a two-port transmission line model with effective components RLGC model [9, 24, 25, 27] as shown in Fig. 2(c). The resistance, conductance, inductance, and capacitance per unit length are

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represented by the symbols R [Ω/m], G [S/m], L [H/m], and C [F/m], respectively.

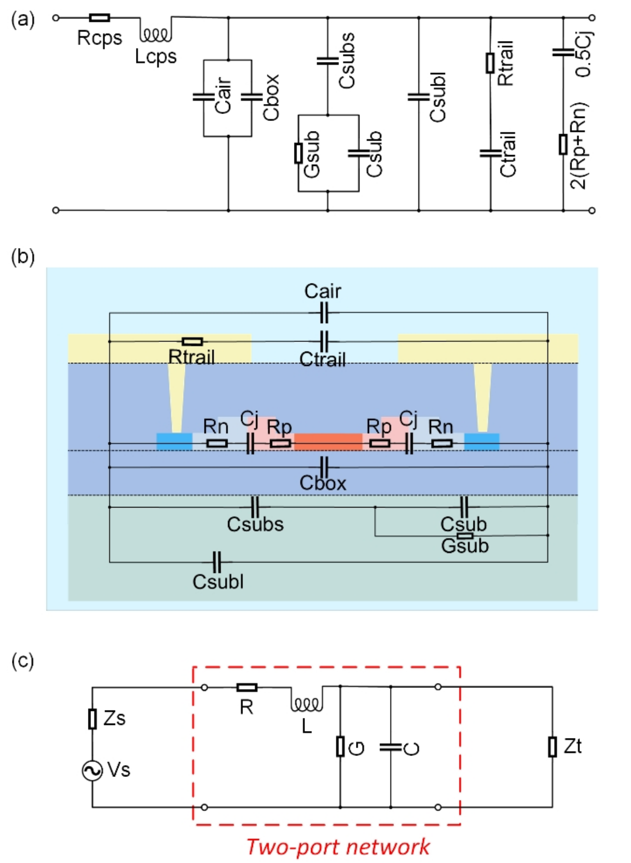

Fig. 2. (a). Equivalent circuit model of the TSWE based EO modulator. (b) Relationship between the circuit and the physical device. (c). Equivalent RLGC model for unit transmission lines.

According to the transmission line model, the capacitance is inferred from the electric field created by the potential difference between the transmission line's two conductors. The inductance represents the generation of a magnetic field due to the current flow in the line. The resistance serves as a representation of the conductor loss. The conductance is used to model the loss in the dielectric between the conductors. The characteristic impedance Z is expressed as:

$$Z=\sqrt{\frac{R+jwL}{G+jwC}}. \tag{1}$$

The complex propagation constant γ is:

$$\gamma=\alpha+j\beta=\sqrt{(R+jwL)\times(G+jwC)}. \tag{2}$$

The radio frequency (RF) index n_{RF} is:

$$n_{RF}=c_0\sqrt{LC}, \tag{3}$$

where c_0 is the speed of light. Three major concerns must be considered to maximize the electro-optical bandwidth of the traveling wave modulator. First of all, microwave attenuation should be as small as possible. Second, the group velocity of the optical and microwave waves must be matched. Research on silicon electro-optic modulators was founded on the assumption that microwave signals propagate in TEM mode without dispersion, that is, that the effective refractive index and the microwave group velocity are equal [29]. Hence, the group velocity of light and the microwave index n_{RF} are required to be matched. Third, the transmission line characteristic impedance should be equal to or greater than the source impedance Z_s and terminal impedance Z_t. Generally, in the conventional microwave communication system, source impedance Z_s and terminal impedance Z_t are 50 Ω. In order to reduce reflections that may induce inter-symbol interference and achieve the largest feasible voltage drop at both ends of the modulator. The electrode structure can be split into two portions to calculate capacitance. A conformal mapping method can be used to perform part of the capacitance calculation for asymmetric coplanar strips (ACPS) with a gap of S_1 and widths of W_{s1} and W_{g1}. The other component is the T-shaped structure's capacitance C_{trail}, which can be computed easily using the plate capacitance formula. We first calculate the capacitance of the transmission line with W_{s1}, W_{g1}, and S_1. Considering the metallization caused by electrode thickness, some authors suggest including the effect of electrode thickness in a similar way to that used for microstrip [30]. The effect of conductor thickness can be modeled by an effective increment of conductor width and a reduction of the gap. We can write the gap as:

$$S_{eff}=S_1-\Delta. \tag{4}$$

Therefore, the effective widths of the source and the ground electrode are:

$$W_S=W_{s1}+\Delta, \tag{5}$$

$$W_G=W_{g1}+\Delta, \tag{6}$$

where Δ accounts for the effect of metallization thickness. The expression for Δ can be expressed by [30, 31]:

$$\Delta=\frac{t}{2\pi\varepsilon_r}\left(1+\ln\frac{8\pi S_1}{t}\right), \tag{7}$$

where t is the thickness of the electrode, ε_r is the relative permittivity of the metal. We can determine the capacitance associated with each layer using the partial-capacitance method and conformal mapping. Fig. 3 illustrates the conformal mapping for the CPS. The geometry of t-plane can be mapped to the w-plane by using the mapping function as follows:

$$w=\int_{t_0}^{t}\frac{dt}{\sqrt{(t-1)(t-t_1)(t-t_2)(t-t_3)}}. \tag{8}$$

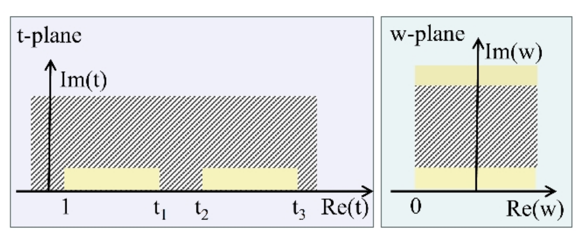

Fig. 3. Coplanar plate capacitor in t-plane and transformed parallel capacitor using conformal mapping in w-plane.

By conformal mapping, the calculation of CPS capacitance can be transformed into the calculation of parallel plate capacitance, and then the capacitance expression of CPS can be obtained through inverse transformation [32]. The electrode plate is in a multi-layer medium. The electric field lines of CPS are distributed in each layer. The capacitance of each dielectric layer is obtained by adding or subtracting the capacitance in different dielectric layers separately [33, 34]. The following definition applies to the geometrical factor for a conformally mapped air-filled CPS:

$$F\left(k_i\right)=\frac{K\left(k_i\right)}{K\left(\sqrt{1-k_i^2}\right)}. \tag{9}$$

where $K(k_i)$ is the complete elliptic integral of the first kind. When i=0, k_0 is given by:

$$k_0=\sqrt{1-\left(\frac{S_{eff}}{W_S+W_G+S_{eff}}\right)^2}, i=0. \tag{10}$$

When i>0, k_i is given by:

$$k_i=\sqrt{1-\left(\frac{\sinh^2\left(\frac{\pi S_{eff}}{4h_i}\right)}{\sinh^2\left(\frac{\pi\left(W_S+W_G+S_{eff}\right)}{4h_i}\right)}\right)^2}, i>0 \tag{11}$$

where h_i is the thickness label in Fig. 1(c). Based on the partial-capacitance technique and the conformal mapping, we can derive the capacitance of the asymmetric coplanar strip line (ACPS) associated with each layer. We use the derivation in [25, 35] to calculate the capacitances. The capacitance C_{air} for the air space above the device can be written as:

$$C_{air}=\varepsilon_0 F^2\left(k_0\right)-\frac{1}{2}\varepsilon_0\frac{F^2\left(k_0\right)}{F\left(k_1\right)}, \tag{12}$$

where ε_0 is the permittivity in vacuum. Analogically, the capacitance of the buried oxide (BOX) layer C_{box} is given by:

$$C_{box}=\frac{1}{2}\varepsilon_0\varepsilon_{ox}\left(\frac{F^2\left(k_0\right)}{F\left(k_2\right)}-\frac{F^2\left(k_0\right)}{F\left(k_3\right)}\right), \tag{13}$$

where ε_{ox} is the silicon dioxide relative permittivity. Contrary to the thick BOX layer, the silicon substrate is a semiconducting layer with nonzero conductivity. This dielectric layer should be described by C_{sub}, C_{subs}, C_{subl}, and G_{sub} as shown in Fig. 2(b). The transverse conductive and displacement currents in the silicon substrate are represented by the capacitor C_{sub} and the conductor G_{sub}, respectively. These two parameters can be written as:

$$C_{sub}=\frac{1}{2}\varepsilon_0\varepsilon_{si}\left(\frac{F^2\left(k_0\right)}{F\left(k_1\right)}-\frac{F^2\left(k_0\right)}{F\left(k_2\right)}\right), \tag{14}$$

$$G_{sub}=2\sigma_{sub}\left(\frac{F^2\left(k_0\right)}{F\left(k_1\right)}-\frac{F^2\left(k_0\right)}{F\left(k_2\right)}\right), \tag{15}$$

where σ_{sub} is the conductivity of the silicon substrate, and ε_{si} is the relative permittivity of the silicon substrate. The capacitance between the signal metal strip and the silicon substrate is represented by the capacitor C_{subs}, which is given by:

$$C_{subs}=\varepsilon_0\varepsilon_{ox}\frac{W_{s1}}{h_2}. \tag{16}$$

In the high-frequency range, an extra capacitor C_{sub1} is present to cause the overall capacitance to converge to C_{sub}. As a result, we have the expression as follows:

$$C_{sub}=\frac{C_{sub}C_{subs}}{C_{sub}+C_{subs}}+C_{subl}. \tag{17}$$

The T-shape electrode is equivalent to the CPS transmission line. The same method can be used to calculate capacitance. Fringe capacitance is considered when calculating the capacitance of the CPS transmission line. However, the capacitance caused by the T-shaped load is relatively small, so the fringe capacitance can be ignored. C_{trail} can be directly calculated using the flat plate capacitance calculation formula, and the expression is as follows:

$$C_{trail}=\frac{\varepsilon_0 t L_{T1}}{S_2\left(L_{T1}+L_{T2}\right)}. \tag{18}$$

After measuring the capacitance, it is required to figure out the transmission line's resistance and inductance. In the low-frequency range, the line resistance can be calculated directly by Ohm's law, while the line conductance can be derived from common magneto-static theory. As the frequency increases, they become frequency-dependent due to the skin effect of the imperfect metal. When there is an alternating current or electromagnetic field in a conductor, the current distribution inside the conductor is uneven. The current density is the largest near the surface of the conductor. We simulated the electrode using HFSS and demonstrated the skin effect in the electrode in Fig. 4(a). We can see the strong current volume distribution is caused by the skin effect. As such, the resistance of the conductor increases. Fig. 4(b) shows the PN junction unloaded cross-section view of the electric field distribution corresponding to Fig.1 (c). The T-rail electrode's electric field distribution is a non-uniform electric field pointing from the signal to the ground.

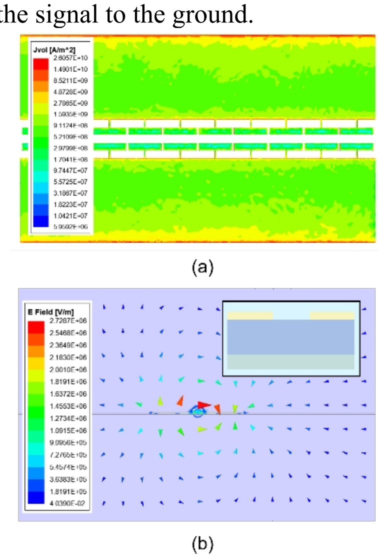

Fig. 4. (a). Top view of current volume distribution in the electrodes at 40 GHz. (b) Cross-section view of the electric field distribution corresponding to Fig.1 (c) without PN junction.

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The skin depth δ is defined as the depth where current density drops to 1/e of the value at the conductor's surface and is given by:

$$\delta = \sqrt{\frac{2}{\omega\mu_0\mu_r\sigma}}. \tag{19}$$

With ω as the angular frequency in radians, μ_0 as the permeability of the vacuum, μ_r as the relative permeability of the conductor, and σ as the conductivity of the electrode. For the CPS electrode with W_{s1}, W_{g1} and S_1. The resistance R_{cps} is the sum of R_c and R_g. R_c is the series resistance in ohms per unit length of the source strip conductor and can be measured by [31]:

$$R_c = \frac{R_s}{2S_{eff}\left(1-k_0^{\,2}\right)\mathrm{K}^2\left(k_0\right)}\left(\pi + \ln\left(\frac{4\pi S_{eff}}{t}\right) - k_0\ln\left(\frac{1+k_0}{1-k_0}\right)\right). \tag{20}$$

The term R_S is the skin effect surface resistance in ohms and is given by:

$$R_s = \frac{1}{\delta\sigma}. \tag{21}$$

R_g is the distributed series resistance in ohms per unit length of the ground planes and is given by:

$$\begin{aligned} R_g &= \frac{k_0 R_s}{2S_{eff}\left(1-k_0^{\,2}\right)\mathrm{K}^2\left(k_0\right)} \\ &\times\left(\pi + \ln\left(\frac{4\pi(S_{eff}+W_{s1}+W_{g1})}{t}\right) - \frac{1}{k_0}\ln\left(\frac{1+k_0}{1-k_0}\right)\right) \end{aligned} \tag{22}$$

As the capacitance calculation process mentioned above, the T-track can be considered as the independent CPS transmission line when calculating the resistance. By incorporating w_{ts}, w_{tg}, and S_2 into formulas 20-22, the resistance R_{trail} introduced by the T-track can be calculated. The inductance per unit length of the T-rail slow-wave transmission line L_{cps} is equal to the inductance of the electrode with W_{s1}, W_{g1}, and S_1 [8, 18, 26]. In the skin-effect region, L_{cps} can be divided into the internal inductance L_{int} and the external inductance L_{ext}. The external inductance is frequency independent, which can be obtained from the capacitance of an air-filled C_{air} and is given by:

$$L_{ext} = \frac{1}{c^2 C_{air}}. \tag{23}$$

The internal inductance can be calculated by Wheeler's incremental inductance rule [26]. Whether the transmission line is symmetrical or not, the internal inductance can be calculated by:

$$L_{\mathrm{int}} = \frac{\delta S_{eff}}{2\varepsilon_0 c^2\left(2F\left(k_0\right)S_{eff}+t\right)^2}\left(\eta\left(\begin{aligned}&\frac{2S_{eff}\left(S_{eff}+W_{s1}\right)}{W_{s1}(W_{s1}+2S_{eff})}\\&-\frac{S_{eff}/2}{W_{s1}+2S_{eff}+W_{g1}}\\&+\frac{S_{eff}/2}{S_{eff}+W_{g1}}\end{aligned}\right)+\left(\frac{t}{S_{eff}}+1\right)\right), \tag{24}$$

where term η can be calculated by the formulas given in [27]. All circuit components in Fig. 2(a) except for the PN junction have been determined now. Fig. 5 shows the variation of RLGC in the T-rail slow-wave electrodes as a function of W_{s1}, W_{g1}, and S_1. Resistance and inductance decrease while conductance and capacitance rise when Ws1 and Wg1 increase. Resistance, conductance, and capacitance all decrease, and inductance rises when S1 increases. Certain discrete sections exhibit uniform responses due to the nonlinear variations in the conductor's charge generated by the skin effect.

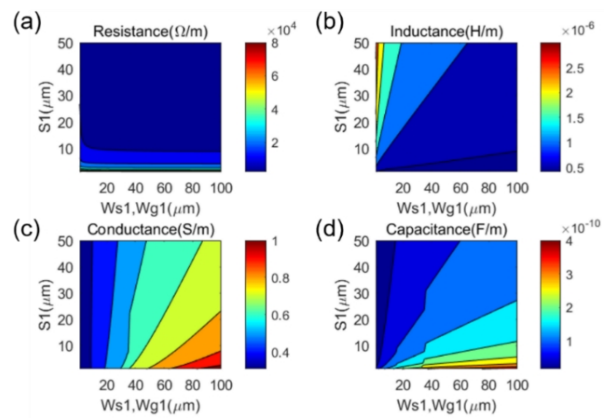

Fig. 5. Simulated per unit length (a) Resistance, (b) Inductance, (c) Conductance, and (d) Capacitance of p-n junction unloaded slow-wave transmission line at 30 GHz. (W_{ts}=W_{tg}=10μm, W_r =2μm, L_{T1}=47μm, L_{T2}=3μm).

If no additional circuit elements are added, the properties during microwave propagation can be calculated from the electrode parameters obtained now. As the PN junction can cause changes in capacitance and resistance values, the overall performance parameters must be combined with the part of the PN junction. Silicon-based EO modulators require capacitance and resistance from the PN junction to obtain microwave propagation data [8, 22].

B. PN junction

We use a CHARGE solver to simulate the carrier concentration as a function of the voltage at the PN junction. Fig. 6. shows the change in n-type carrier concentration at various reverse voltages of the PN junction. Increasing the reverse bias voltage widens the depletion region and affects the resistance and capacitance of the PN junction, thereby impacting the modulator performance.

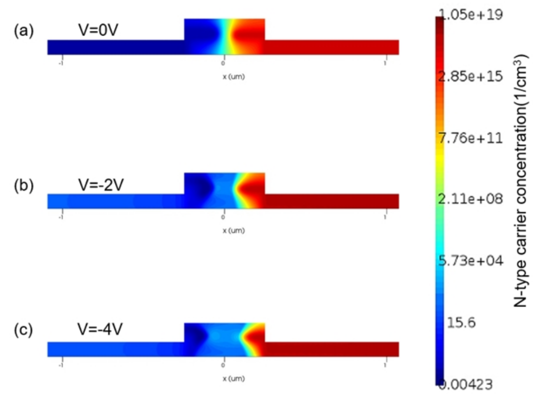

Fig. 6. Cross-sections of the n-type carrier concentration of the PN junction under a bias voltage of 0V, 2V, and 4V. P-type doping is found on the left side, while n-type doping is seen on the right.

The resistance of the PN junction mainly depends on the doping concentration. The approximate value of resistance and capacitance of the PN junction can be obtained by assuming uniform doping of the PN junction. The resistance introduced by the PN junction can be calculated from the square resistance and volume of each region. The resistance of the PN junction R_p and R_n are in series, the resistance can be calculated by [36]:

$$\begin{aligned} R_j &= R_p + R_n \\ &= \left(\frac{w_{rib}}{2} - w_p\right) R_{srp} + \left(\frac{w_{rib}}{2} - w_n\right) R_{srn} \\ &+ \left(x_{p++} - \frac{w_{rib}}{2}\right) R_{ssp} + \left(x_{n++} - \frac{w_{rib}}{2}\right) R_{ssn} \end{aligned} \tag{25}$$

where w_{rib} is rib width, t_{rib} is rib height, and R_{srn}, R_{srp}, R_{ssn}, and R_{ssp} are the sheet resistances of the n-doped rib, p-doped rib, n-doped slab, and p-doped slab, respectively. With

$$w_n = w_p = \frac{W_d}{1 + N_A / N_D}. \tag{26}$$

W_d is determined by the impurity densities (N_A for P doped area and N_D for N doped area), as well as the applied voltage V, and is given by:

$$W_d = \sqrt{\frac{2\varepsilon_0 \varepsilon_s \left(N_A + N_D\right)\left(V_{bi} - V\right)}{q N_A N_D}}. \tag{27}$$

The capacitance of the p-n junction is given by:

$$C_j = t_{rib} \sqrt{\frac{q\varepsilon_0 \varepsilon_s}{2\left(\frac{1}{N_D} + \frac{1}{N_A}\right)\left(V_{bi} - V\right)}}, \tag{28}$$

where V_{bi} is the built-in or diffusion potential of the junction given by:

$$V_{bi} = \frac{k_B T_t}{q} \ln \frac{N_A N_D}{n_i^2}, \tag{29}$$

where n_i is intrinsic charge carriers in m^{-3}, k_B is Boltzmann constant in J/K, T_t is the temperature in K. The resistance and capacitance of PN junctions of various sizes at two doping levels are depicted in Figs. 7(a) and 7(b) respectively. An increase in the bias voltage causes the capacitance to increase and the resistance to slightly decrease.

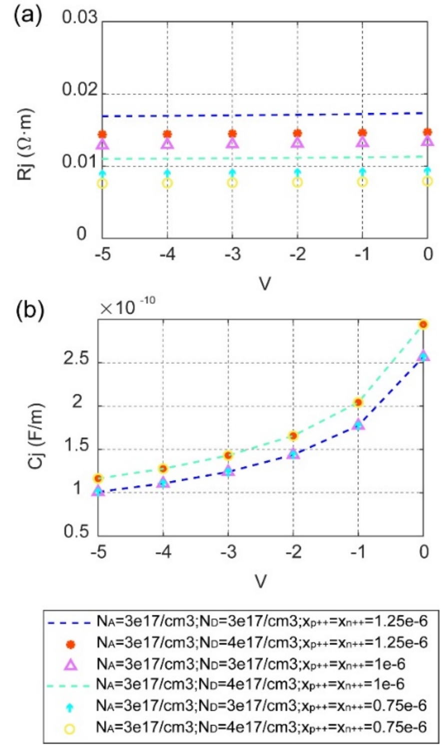

Fig. 7. (a) Resistance of the lateral p-n junction vs. voltage. (b) The capacitance of the lateral p-n junction vs. voltage.

The transmission line's overall performance parameters can be computed by adding the electrode's RLGC parameters to the capacitance and resistance produced by the PN junction. The loss will increase due to the resistance produced by the PN junction. The intermediate doping layer can be added to reduce resistance and thereby reduce microwave losses. If the PN junction is too narrow, there may be numerous electrode design restrictions that could compromise the device's overall performance [15]. With a larger PN junction, it is possible to gain high design flexibility for the electrodes, but the larger doping zone also results in more optical loss. Tradeoffs must be made during the design process.

III. SILICON OPTICAL MODULATORS DESCRIPTION AND ANALYSIS

The electrode design parameters from Ref [8] were used to verify the accuracy of our equivalent circuit. We employed our equivalent circuit model and HFSS software to compute the primary parameters of the t-shaped electrode. The outcomes of our calculations are shown in Fig. 8. The data suggests that the results obtained from our circuit model are consistent with the outcomes obtained from HFSS. Fig. 8(a) depicts the microwave loss feature of the transmission line. In this instance, the primary source of loss is usually determined to be conductor loss rather than dielectric loss. The microwave index of the modulator can be seen in Fig. 7(b), which is close

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to the optical group index in the silicon waveguide (3.8-4) [22]. Characteristic impedance Z_0 results for the reference modulator against frequency for the equivalent circuit model approach and the HFSS simulator are shown in Fig. 8(c) showing that, as expected, Z_0 decreases as the frequency is increased due to the skin effect [28]. The characteristic impedances are extremely close to 50Ω, which is the optimal state for microwave systems. The circuit model maintains an accuracy of 2% when compared to the HFSS simulation. Additionally, as pointed out before, the quasi-static approximation can maintain the deviation of capacitance less than 6% when the frequency is less than 100GHz [33].

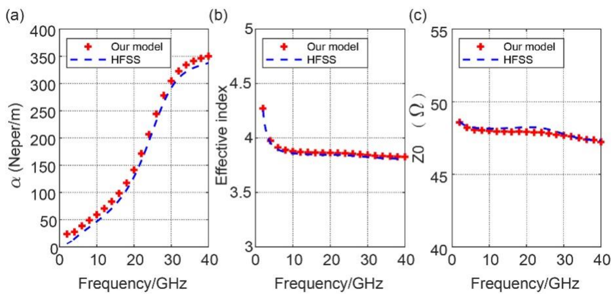

Fig. 8. (a) Microwave loss, (b) RF index, and (c) Characteristic impedance calculated by our model and HFSS simulator. (W_{s1}=W_{g1}=120μm, S_1=51μm, S_2=12.6μm, W_{ts}=W_{tg}=10μm, W_r=2μm, L_{T1}=47μm, L_{T2}=3μm, L_{eff}=4mm , x_{p++}=x_{n++}=1.25μm).

The modulator's small signal electro-optic response as a function of velocity mismatch and total microwave loss can be calculated by [14]:

$$M(f)=e^{\frac{-\alpha_0\sqrt{f}L_{eff}}{2}}\times\left(\frac{\sinh^2\left(\frac{\alpha_0\sqrt{f}L_{eff}}{2}\right)+\sin^2\left(\frac{\pi fL_{eff}\times\left(n_{RF}-n_{opt}\right)}{c}\right)}{\left(\frac{\alpha_0\sqrt{f}L_{eff}}{2}\right)^2+\left(\frac{\pi fL_{eff}\times\left(n_{RF}-n_{opt}\right)}{c}\right)^2}\right)^{\frac{1}{2}}, \quad (30)$$

where α_0 is the field attenuation coefficient, n_{RF} is the effective refractive index of the microwave, n_{opt} is the group effective refractive index of the optical wave, L_{eff} is the effective length respectively, with

$$L_{eff}=\frac{LL_{T1}}{L_{T1}+L_{T2}}. \quad (31)$$

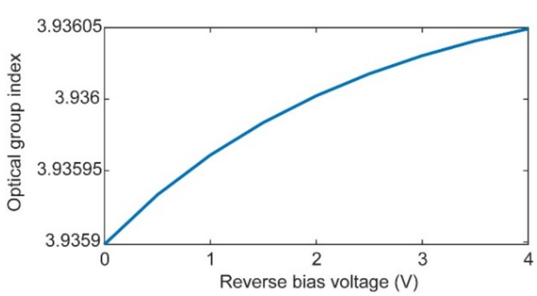

Fig. 9. The simulated optical group index as a function of the reverse bias voltage.

The optical group index in silicon-based modulators is typically 3.8-4 and is influenced by factors like doping concentration in the PN junction [22]. We used the CHARGE solver and the MODE solver to simulate the optical group index at various reverse bias voltages and presented the results in Fig. 9. The group refractive index is around 3.9. The response of the EO modulator, as calculated by the use of HFSS and the equivalent circuit model, is depicted in Fig. 10. The material of the electrode is aluminum. The electrode design parameters are: W_{s1} = W_{g1} = 120 μm, S_1 = 51 μm, W_{ts} = W_{tg} = 10 μm, g = 9.2 μm, W_r = 2 μm, L_{T1} = 47 μm, L_{T2} = 3 μm. The parameters of PN junction are $N_A = N_D = 4\times10^{17}/cm^3$, x_{n++} = x_{p++} = 1.2 μm. The increase in bias voltage given to the PN junction leads to an observed increase in the bandwidth of the electro-optic modulator, as demonstrated by both calculation approaches. The EO response estimated using the equivalent circuit model, exhibits similarities to its HFSS counterpart.

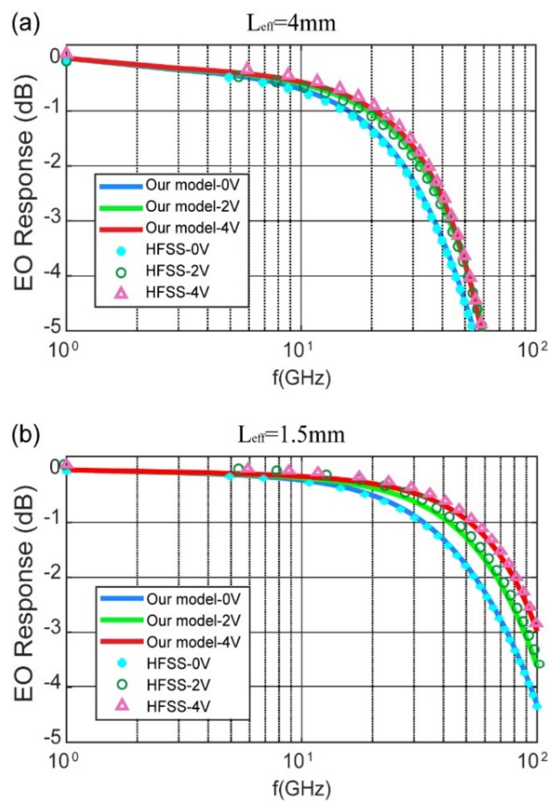

Fig. 10. (a) EO frequency responses using equivalent circuit model and HFSS under different bias voltages when the effective length is 4mm. (b) EO frequency responses using equivalent circuit model and HFSS under different bias voltages when the effective length is 1.5mm.

Fig. 10(a) shows that the 4mm effective long modulator provides 43-GHz bandwidth. We further model a modulator with 100-GHz bandwidth by setting the effective length to 1.5 mm in our proposed method. The simulation results of the proposed model and the HFSS are shown in Fig. 10(b). The simulation results of the proposed model are highly consistent to the result of the HFSS.

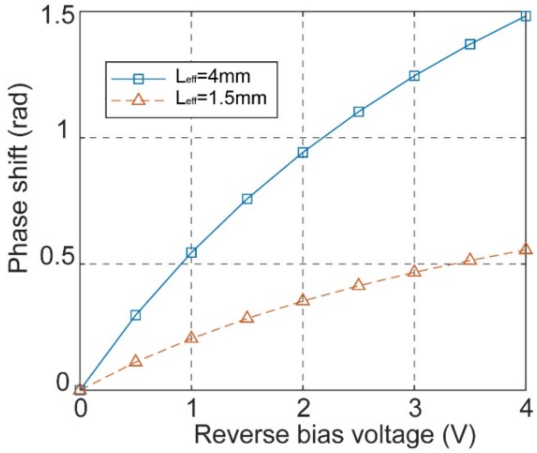

Fig. 11 The simulated phase shift as a function of the reverse bias voltage.

The simulated phase shift as a function of the reverse bias voltage is illustrate in Fig. 11. Theses result translate to a DC Vπ of 5.76V and 15.36V for the 4mm long and 1.5mm long modulator respectively at a bias of −1 V. The efficiency decreased as the depletion region area become reduced due to the increase in reverse bias [16]. When biased at -2V, -3V, and -4V, the DC Vπ of the 4mm effective long device are 6.67V, 7.57V, and 8.48V respectively. Meanwhile, the DC Vπ of the 1.5mm effective long device is 17.79V, 20.19V, and 22.61V respectively.

IV. Conclusions

The equivalent circuit model of the carrier-depletion-based silicon modulators with slow-wave transmission lines is presented. The circuit model of the T-shaped electrode is commonly used in silicon photonics modulators, III-V compound semiconductor electro-optic modulators, and other traveling wave devices. Our equivalent circuit model can accurately estimate key parameters related to TSWE. The acquired results are well-matched to the HFSS simulation results. We concluded that these formulas serve as an efficient design and optimization tool in various applications due to their correctness, simplicity, and ability to handle a wide range of T-rail electrode structures.

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